

STOKE PREDICTION

Group 15

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OBJECTIVES

- ➔ Stroke is one of the leading causes of death and disability worldwide. Early prediction of stroke can help doctors take preventive actions and improve patient outcomes.
- ➔ Used healthcare data and machine learning techniques to analyze patient health information.
- ➔ The goal is to identify individuals who are at high risk of stroke based on factors such as age, BMI, glucose level, and medical history.

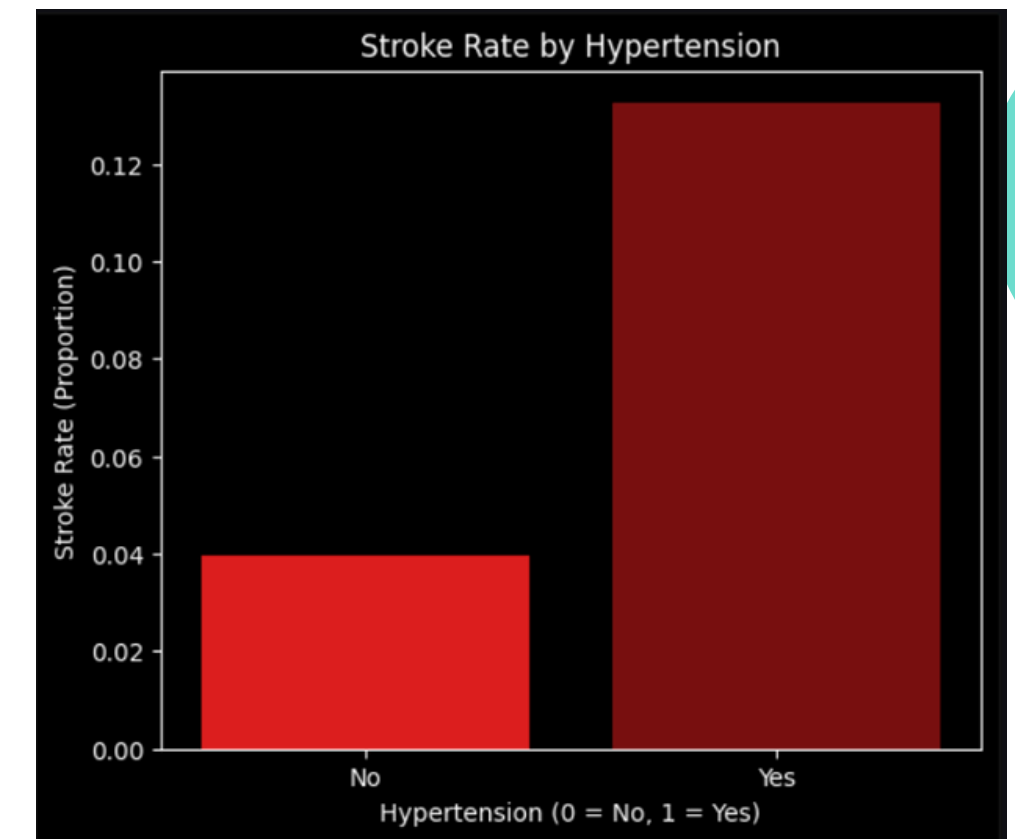
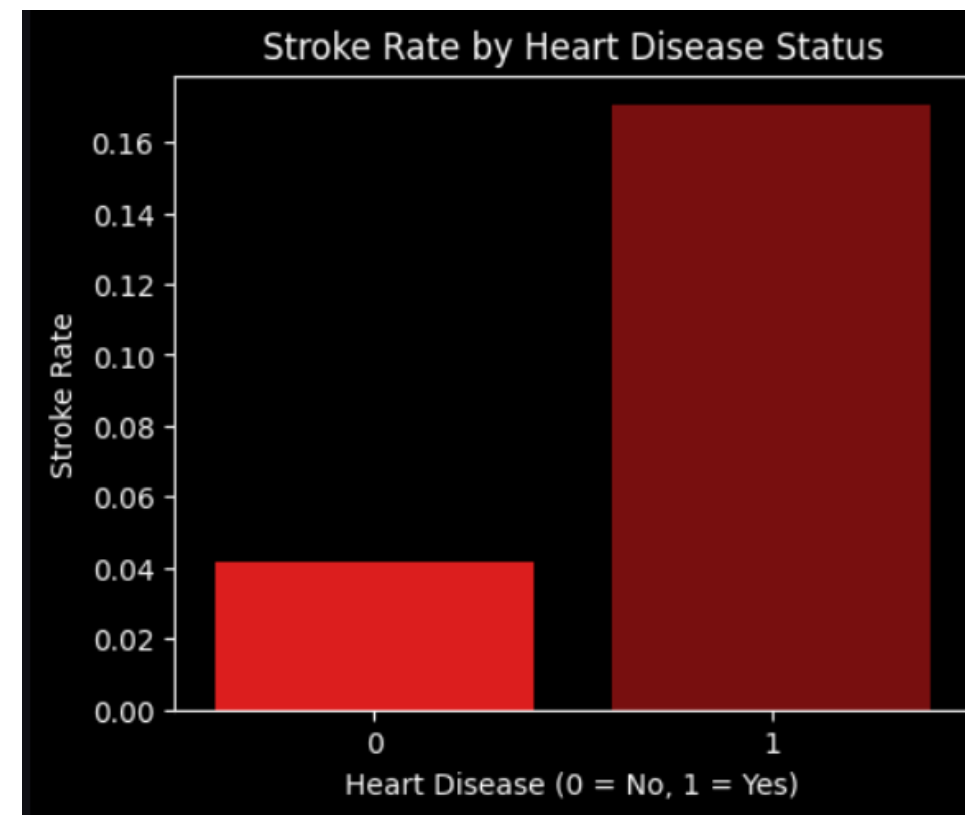
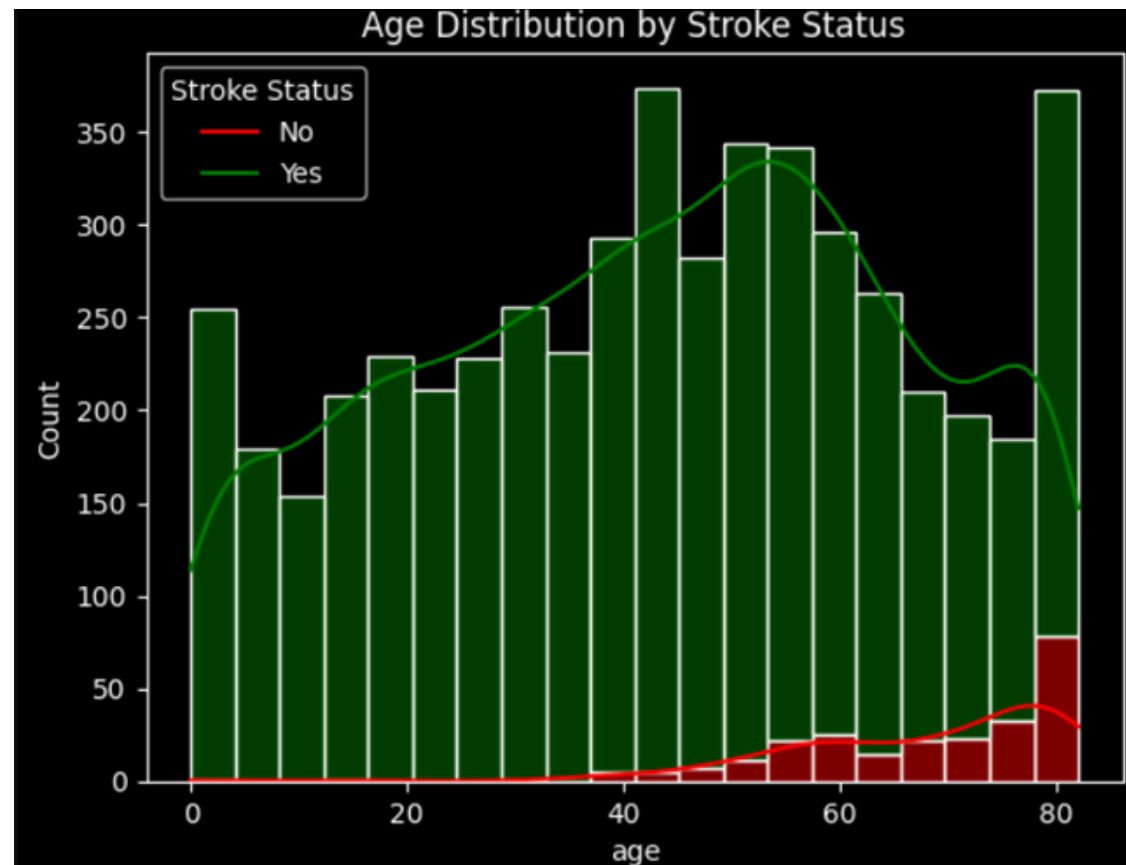


DATA DESCRIPTION

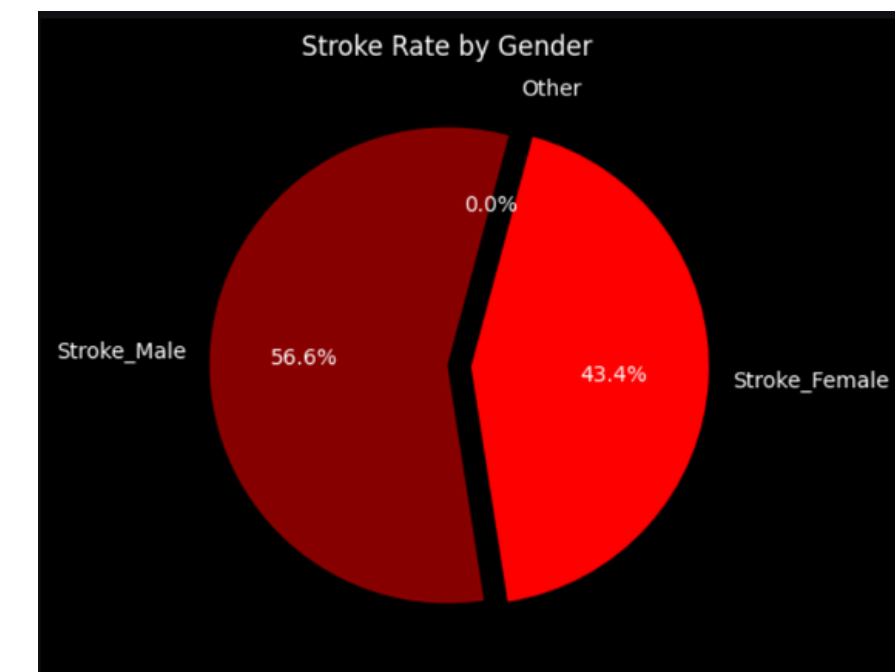
- ① Data preprocessing was performed to clean the dataset and handle missing values, especially in the BMI column.
- ② Total records: 5,110 patients, Stroke cases: 249, Non-stroke cases: 4,861
- ③ Categorical variables were converted into numerical form using Label encoding techniques.
- ④ Machine learning models such as Logistic Regression and Random Forest were used to predict stroke risk.
- ⑤ Applied feature scaling using StandardScaler.



EXPLORATORY DATA ANALYSIS



- Stroke patients are generally old
- Hypertension and heart disease increase risk.
- Stroke_Male: 56.6%
- Stroke_Female: 43.4%



MACHINE LEARNING MODELS

MODELS	Precision	Recall	F1-Score	Accuracy
Logistic Regression	0.17	0.50	0.25	0.85
Random Forest	0.19	0.22	0.29	0.92
XgBoost	0.17	0.34	0.23	0.89

Random Forest is the best-performing model due to its highest accuracy and F1-score



HYPOTHESIS TESTING



Chi-Square Test

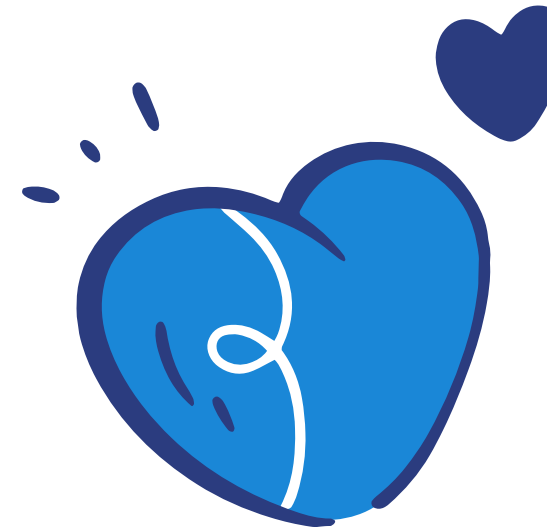
- **Hypertension, heart disease, smoking status, work type, and marital status** showed a significant association with stroke ($p\text{-value} < 0.05$).
- **Gender and residence type** were found to be not significantly associated with stroke ($p\text{-value} \geq 0.05$).
- **2 Independent Samples**
 - Group 1: Patients with stroke
 - Group 2: Patients without stroke



HYPOTHESIS TESTING

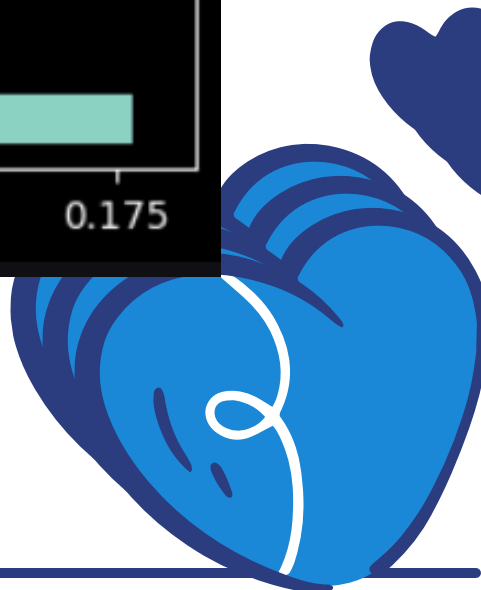
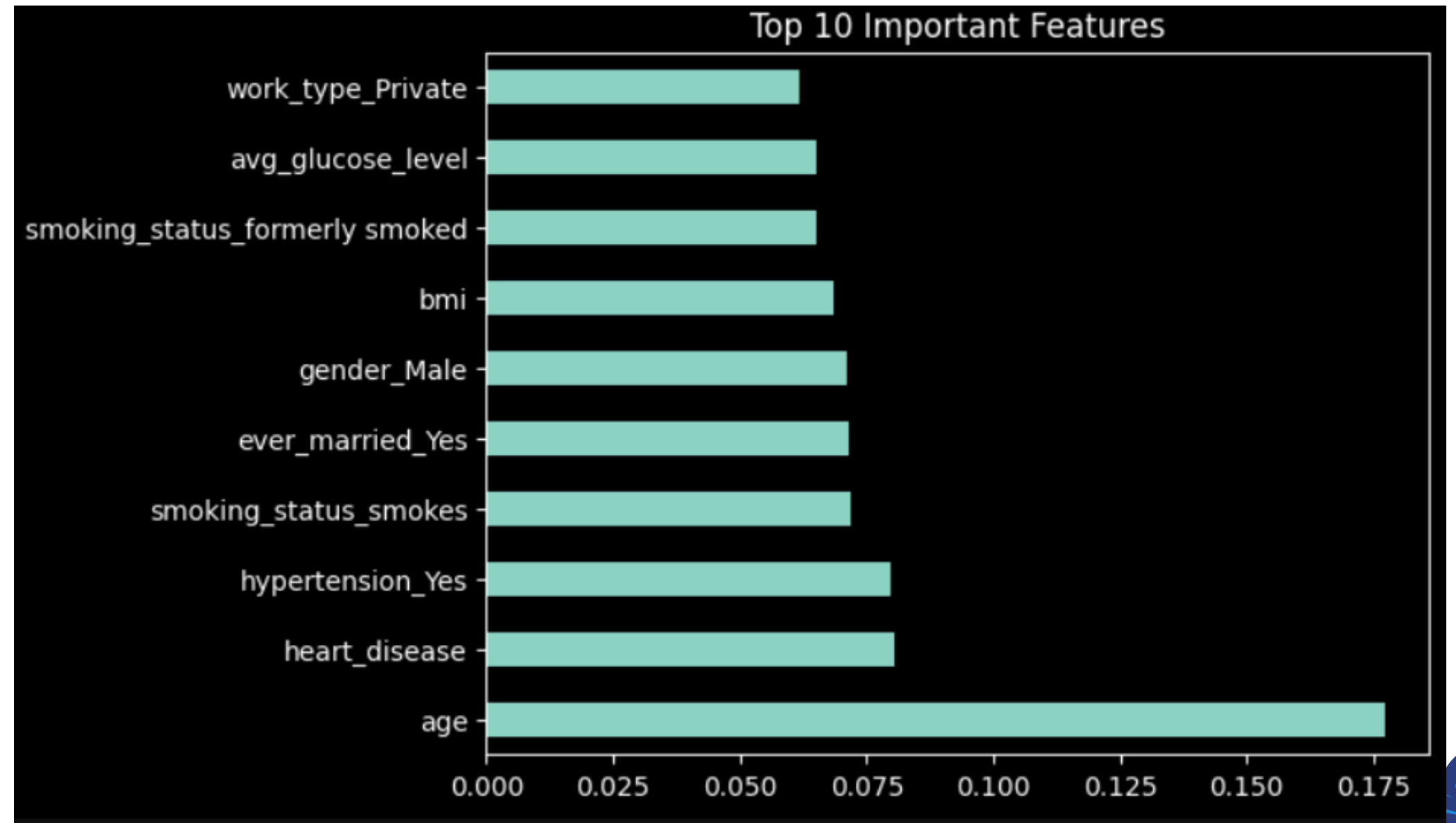
➡ T-test

- Age, BMI, and glucose levels show statistically significant differences between stroke and non-stroke patients.
- These variables are important predictors for stroke risk in the machine learning model.



FEATURE IMPORTANCE

Age is the dominant factor of predictor of stroke, followed by heart disease, hypertension, smoking, BMI, and glucose level.



Age

45

Hypertension

No



Heart Disease

Yes



Ever Married

Yes



Work Type

Deployment Model

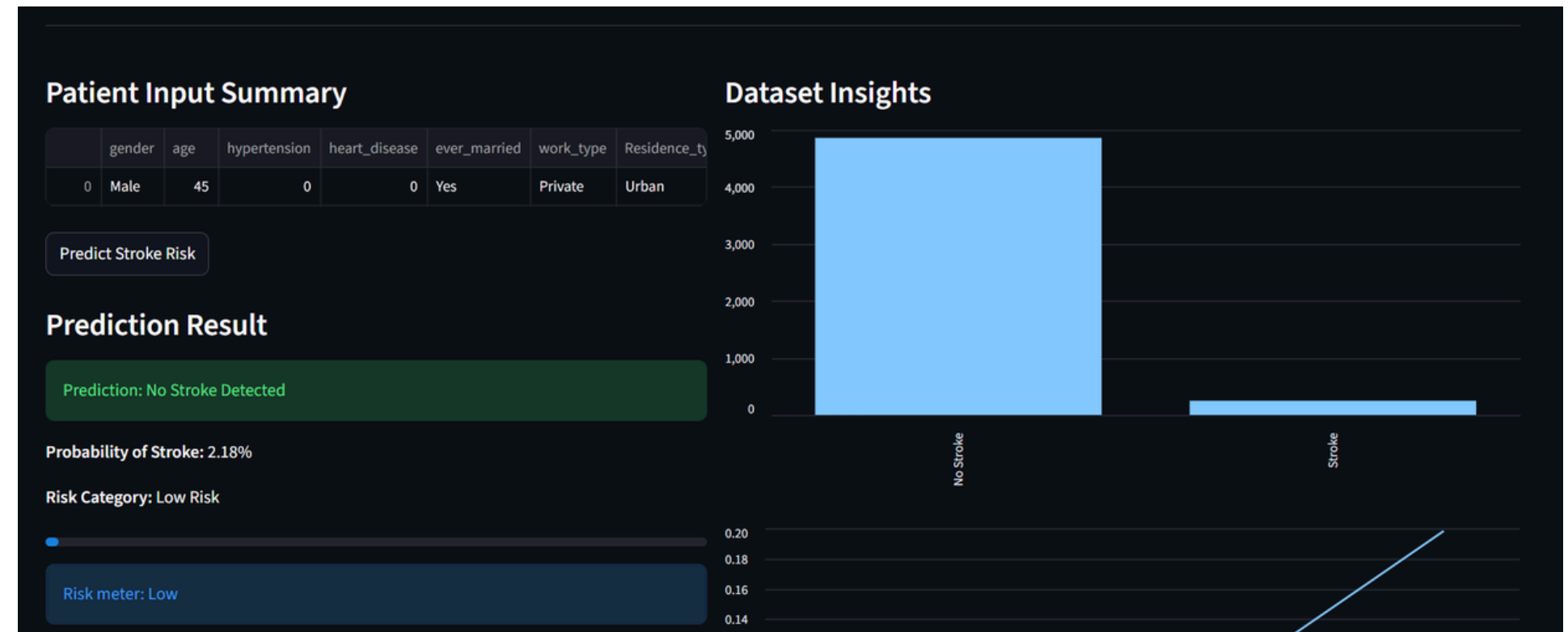
Patient Details

- Using patient health details like age, heart disease, work type, gender, ever married and other factors
- Use Streamlit app



Deployment Model

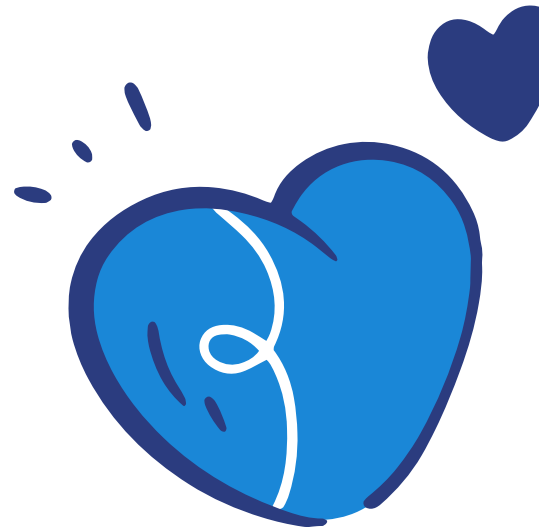
- Predict Stroke on patient 'health factor
- Best Model is Random forest to predict the stroke rate





CONCLUSION

Machine learning algorithms were used for prediction of the risk of strokes based on the healthcare data. Hypothesis testing proved useful in determining significant factors like age, high blood pressure, and glucose level. Random Forest model is analyzed by measuring accuracy and confusion matrix to predict the stroke Rate. Deployment of the model was done via Streamlit.





**THANK YOU
SO MUCH** 